FIRST TECH CHALLENGE | TEAM #31113

ROBOLADS · ENGINEERING PORTFOLIO

Season: Decode (2025-26) / Into The Deep (2024-25) | Location: Hanoi, Vietnam
Organization / School: High School for Gifted Students (HSGS), VNU Hanoi

MAJOR AWARDS & JUDGING RECOGNITION

· Inspire Award Winner · FIRST Tech Challenge Vietnam Championship
· Innovate Award Winner · Sponsored by Raytheon Technologies
· Think Award Finalist · Bilingual Engineering Documentation & Rigor

EXECUTIVE SUMMARY

HSGS RoboLads (FTC #31113) is a premier Vietnamese student robotics unit dedicated to competitive robotics excellence and open-source STEM education. Our portfolio documents the iterative design of Cerberus, from initial napkin sketches to competitive tournament execution.

CORE SUBSYSTEMS & MECHANISMS

· Active Coaxial Continuous Intake
· 4-Stage Carbon Fiber / Aluminium Cascade Lift
· Virtual 4-Bar High-Torque Specimen Manipulator
· Custom 3-Wheel Dead-Wheel Odometry Pods

MECHANICAL & CAD ENGINEERING HIGHLIGHTS

· Active Coaxial Roller Intake: Continuous intake system utilizing compliant silicone rollers, eliminating jamming during rapid intake transitions under defense.
· 4-Stage Cascade Linear Lift: CNC-machined 4mm 6061 aluminium plates paired with carbon fiber tubes for high torsional rigidity under 11 kg total robot weight.
· Virtual 4-Bar Specimen Gripper: Custom linkage articulated by coreless digital servos with 3D-printed TPU 95A compliant gripper pads.
· Modular Electronics Bay: Slide-out Control Hub tray with Anderson Powerpole quick-disconnect harnesses allowing rapid 60-second module replacements.

AUTONOMOUS NAVIGATION & SOFTWARE

· RoadRunner v1.0 Motion Profiling: 3-wheel dead-wheel odometry pods achieving sub-0.5 inch position repeatability.
· AprilTag Vision Pipeline: Real-time pose estimation via Limelight 3G eliminating cumulative odometric drift during multi-cycle autonomous runs.
· Single-Button Macro Sequencing: Automated lift extension and gripper deposit logic synchronized for 1.2-second cycle completions.

COMMUNITY OUTREACH & STEM ADVOCACY

CONNECTING & MOTIVATING INITIATIVES

· Vietnam Open CAD Initiative: Distributed free Onshape STEP files and parametric 3D models to over 15 rookie FTC teams across Southeast Asia.

· High School Robotics Bootcamps: Hosted 8 interactive hands-on workshops for 250+ secondary students across Hanoi, teaching mechanical assembly and Java programming.

· Bilingual FTC Documentation: Authored and published full Vietnamese translations of RoadRunner motion profiling, AprilTag vision processing, and PID control fundamentals.

· Regional Mentorship & Scrimmages: Organized competitive practice matches and technical review sessions for regional FTC teams preparing for championship qualifiers.